Library Management System in C

Program Overview

SYSTEM PURPOSE AND STRUCTURE OF THE LIBRARY MANAGEMENT SYSTEM

This program is a **simple console-based system** designed to efficiently add, view, search, and delete books, providing essential functionalities for library management. The main menu offers clear options for user interactions.

Book Data Structure

```c
struct Book
{
    int id;
    char name[30];
    char author[30];
    float price;
};

struct Book b[100];
int count = 0;

/* Function Declarations */
void addBook();
void viewBooks();
void searchBook();
void deleteBook();
```

STRUCT FOR MANAGING BOOK DETAILS

Core Functions Overview

01

ADD BOOK

Collects and stores book details securely.

02

VIEW BOOKS

Displays all books currently in the system.

03

SEARCH BOOK

Finds a book by ID and displays details.

04

DELETE BOOK

Finds The ID of the book and then deletes it.

Function Deep Dive - Add Book And Delete Book

```c
/* ADD BOOK */
void addBook()
{
    printf("\nEnter Book ID: ");
    scanf("%d", &b[count].id);

    printf("Enter Book Name: ");
    scanf("%s", b[count].name);

    printf("Enter Author Name: ");
    scanf("%s", b[count].author);

    printf("Enter Price: ");
    scanf("%f", &b[count].price);

    count++;

    printf("Book Added Successfully!\n");
}
```

<ADD BOOK>

```c
/* DELETE BOOK */
void deleteBook()
{
    int id, i, j, found = 0;

    printf("Enter Book ID to delete: ");
    scanf("%d", &id);

    for (i = 0; i < count; i++)
    {
        if (b[i].id == id)
        {
            for (j = i; j < count - 1; j++)
            {
                b[j] = b[j + 1];
            }

            count--;
            found = 1;
            printf("Book Deleted Successfully!\n"
            break;
        }
    }
  }

  if (found == 0)
      printf("Book Not Found!\n");
}
```

DELETE BOOK

The **addBook() function** reads user input and stores book details, ensuring accurate record keeping and smooth operation.

View & Search Functions

LOGIC BEHIND BOOK MANAGEMENT

The viewBooks() function prints all stored books or a message if none exist, while searchBook() locates a book by ID through conditional checks.

```c
/* VIEW BOOKS */
void viewBooks()
{
    int i;

    if (count == 0)
    {
        printf("No Books Available!\n");
        return;
    }

    printf("\nID\tName\tAuthor\tPrice\n");

    for (i = 0; i < count; i++)
    {
        printf("%d\t%s\t%s\t%.2f\n",
                b[i].id,
                b[i].name,
                b[i].author,
                b[i].price);
    }
}
```

```c
/* SEARCH BOOK */
void searchBook()
{
    int id, i, found = 0;

    printf("Enter Book ID to search: ");
    scanf("%d", &id);

    for (i = 0; i < count; i++)
    {
        if (b[i].id == id)
        {
            printf("Book Found!\n");
            printf("%d %s %s %.2f\n",
                    b[i].id,
                    b[i].name,
                    b[i].author,
                    b[i].price);

            found = 1;
            break;
        }
    }

    if (found == 0)
        printf("Book Not Found!\n");
}
```

Thank You!